10.2 Solution

First convert time into years:

$$t = \frac{253}{365}.$$

Now substitute into the formula:

$$F = 303\left(1 + 0.01 \cdot \frac{253}{365}\right).$$

Compute the interest adjustment:

$$0.01 \cdot \frac{253}{365} \approx 0.0069315.$$

So,

$$F = 303(1.0069315).$$

Therefore,

$$F \approx 305.1002.$$

Rounded to the nearest cent,

$$\boxed{F = \$305.10.}$$

10.3 Correct Answer

The correct choice is

$$\boxed{305.10.}$$

Remark 10.1. Because the stock pays no dividends in this exercise, the forward price is above the current stock price by the financing cost of carrying the stock to the future date.

11 Options on Listed Futures

11.1 Calls and Puts

Definition 11.1 (Call Option)**.** A **call option** is the right, but not the obligation, to buy the underlying asset at a specified price, called the strike price, on or before expiration.

Definition 11.2 (Put Option)**.** A **put option** is the right, but not the obligation, to sell the underlying asset at a specified strike price on or before expiration.

Remark 11.1. Options are derivative instruments. Their value is based on the value of an underlying contract, often a future or a stock.

11.2 Reading a Basic Options Table

Remark 11.2. A listed options market is usually organized by expiration date and strike. For each strike, the market displays bid, theoretical value, and ask for both calls and puts.

Expiration date = June 20, 2022

Calls				Puts		
bid	theo	ask	strike	bid	theo	ask
7.75	8.00	8.25	315	5.75	6.00	6.25
5.25	5.50	5.75	320	8.25	8.50	8.75

Figure 8: A simplified listed-options table showing calls and puts for two strike prices under a single expiration date.

Example 11.1 (Interpreting the 315 Strike)**.** Suppose the 315 call has a theoretical value of 8.00 and is quoted

$$7.75 \text{ at } 8.25.$$

A buyer paying 8.25 purchases the right to buy the underlying asset at 315. The seller of that option takes on the obligation to sell the underlying at 315 if the option is exercised.

Example 11.2 (Interpreting the 315 Put)**.** Suppose the 315 put has a theoretical value of 6.00 and is quoted

$$5.75 \text{ at } 6.25.$$

A buyer of the put acquires the right to sell the underlying at 315. The seller of the put takes on the obligation to buy the underlying at 315 if the option is exercised.

Remark 11.3. Different strikes are listed one below another. For a given expiration, there are usually many strike prices above and below the current underlying level.

11.3 A More Complete Options Screen

Remark 11.4. In a full options screen, the strikes usually run down the middle, with call markets on one side and put markets on the other. In addition to bid and ask prices, the screen may show theoretical value, bid size, ask size, and trader position columns.

Example 11.3 (Reading One Option Line)**.** Suppose the 55-strike call has:

$$\text{bid} = 1.775, \qquad \text{theo} = 1.85, \qquad \text{ask} = 1.90.$$

This means:

- the highest current bid is 1.775,
- the desk's theoretical value is 1.85,
- the lowest current offer is 1.90.

If the bid quantity is 85 and the ask quantity is 63, then 85 contracts are bid at 1.775 and 63 contracts are offered at 1.90.

Remark 11.5. The bid is the best immediately sellable price. The ask is the best immediately buyable price. The theoretical value sits between them as the market maker's estimate of fair value.

11.4 Looking Up Options on Yahoo Finance

Remark 11.6. Public sites such as Yahoo Finance can be used to inspect listed option chains. While these screens are delayed, they are still useful for learning how strikes, expirations, and bid-ask markets are displayed.

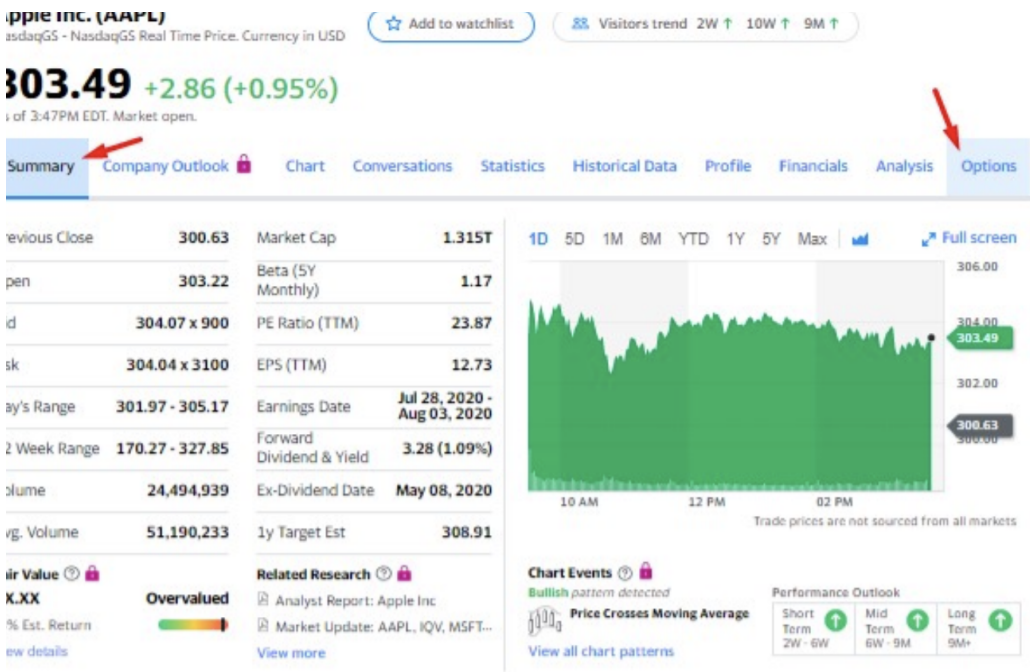

Figure 9: A stock summary page. The options tab leads to the listed option chain for that stock.

Example 11.4 (First Step)**.** To look up options on Apple, one may first search for the ticker `AAPL` on Yahoo Finance and open the stock summary page. From there, clicking the **Options** tab opens the option chain.

Remark 11.7. Once inside the option chain, the user can choose an expiration date and inspect the calls and puts listed for that maturity. A straddle-style view is often more helpful than a simple list because it places calls and puts on the same line by strike.

Example 11.5 (Comparing Calls and Puts Near Spot)**.** Suppose the stock is trading around 303.49 and we inspect the 300 strike. If the call and put values are relatively close, that should not be surprising. Both contracts are near the current stock price, and their relationship will later be explained more formally through put-call relationships.

11.5 Contract Specifications

Definition 11.3 (Contract Specifications)**. Contract specifications** are the exchange-defined details of an option or futures contract, such as expiration date, strike listing conventions, settlement type, tick increment, and multiplier.

Remark 11.8. These specifications vary by product and are always set by the exchange. For options on futures, it is especially important to know which underlying future the option references and how exercise and settlement work.

11.6 Multipliers and Contract Size

Remark 11.9. Financial contracts do not all translate price movement into dollars in the same way. The multiplier converts a move in points into actual cash value.

Definition 11.4 (Multiplier)**.** The **multiplier** is the scaling factor that converts the quoted contract price into notional value and profit-and-loss.

Example 11.6 (Stock Multiplier)**.** If one share of stock is bought at \$175.05 and later sold at \$175.80, then the gain is

$$175.80 - 175.05 = 0.75.$$

Since the stock multiplier is 1 share, the total profit is simply \$0.75 per share.

Remark 11.10. For stocks, this is simple because the multiplier is 1. For futures and options on futures, the multiplier is often much larger and product-specific.

11.7 Futures Multipliers

Example 11.7 (Corn Futures)**.** One corn future represents 5000 bushels of corn. If corn is quoted at 384 cents per bushel, then the notional value of one contract is

$$3.84 \times 5000 = 19{,}200.$$

Remark 11.11. A one-cent move in corn corresponds to

$$0.01 \times 5000 = 50$$

dollars per contract.

Remark 11.12. Corn futures trade in quarter-cent ticks, that is,

$$0.25 \text{ cents} = 0.0025 \text{ dollars}.$$

Since a full cent is worth \$50, one quarter-cent tick is worth

$$\frac{50}{4} = 12.50$$

dollars per contract.

Example 11.8 (Corn Tick P&L)**.** If we buy 10 corn futures at 383.75 and sell 10 at 384.00, then we earn one tick on each contract. The profit is

$$0.25 \times 0.01 \times 5000 \times 10 = 125.$$

So the total profit is

$$\boxed{\$125.00.}$$

11.8 Options Multipliers

Remark 11.13. Options also have multipliers. Stock options usually have a multiplier of 100, meaning one option corresponds to 100 shares. Options on futures usually have a multiplier of 1 option to 1 future, but that future itself may carry a large underlying commodity multiplier.

Example 11.9. A corn option typically converts into one corn future if exercised. Since one corn future represents 5000 bushels, the economic exposure of the option is tied to that futures multiplier.

11.9 Edge Versus Cash

Definition 11.5 (Edge)**.** **Edge** is the amount by which a trade price differs from the trader's current theoretical value, measured in points or ticks.

Remark 11.14. Traders often discuss option trades in terms of edge instead of immediate cash P&L. This is especially common in options trading, where the important first question is often whether the trade was done above or below theoretical value.

Example 11.10 (Edge in Points)**.** Suppose an option is worth 1.00 theoretically and we sell it at 1.04. Then we made

$$1.04 - 1.00 = 0.04$$

points of edge.

Remark 11.15. If the option ticks in pennies, then 0.04 points is also 4 ticks of edge. If the option ticks in 2-cent increments, then the same 0.04 corresponds to 2 ticks.

Definition 11.6 (Cash P&L from Edge)**.** To convert edge into cash, use

$$\text{Cash P\&L} = \text{edge} \times \text{contracts traded} \times \text{multiplier}.$$

Example 11.11 (Edge to Cash)**.** Suppose we sell 20 contracts of an option at 1.04 when our theoretical value is 1.00, and suppose the multiplier is 1000. Then the edge is 0.04, so the corresponding cash P&L is

$$0.04 \times 20 \times 1000 = 800.$$

Thus the trade represents

$$\boxed{\$800}$$

of theoretical profit relative to our current theo.

11.10 Corn Option Edge Example

Example 11.12 (Corn 400 Call)**.** Suppose we sell 20 corn 400 call options at 20.25 and we have them worth 20.00. Then our edge is

$$20.25 - 20.00 = 0.25.$$

Because agricultural options often tick in 0.125 increments, this corresponds to

$$\frac{0.25}{0.125} = 2$$